Stochastic Smooth Discrete Local Volatility

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Abstract

We present a statically and dynamically arbitrage-free European option market model with smooth interpolation between call prices, based on the SANOS [6] framework.

In spirit, it is a stocastic model for local volatility when the latter is used as a representation of a European option price surface, not as a pricing model.

We illustrate the strength of our approach on S&P Index Option data from January 2022 to August 2025. We show that the model's initial state fits within bid/ask for more than 95% of relevant market option prices.[1] We then train the model dynamics under $\mathbb{P}$ and $\mathbb{Q}$ and show that the model is indeed free of arbitrage while fitting the dynamics of the market well.

1 Introduction

This article presents an arbitrage-free market model for for smooth surfaces of European option prices. While not strictly the first of is kind, it is the first model with an industy-grade ability to fit the current market, allow for smooth interpolation of the entire option surface at each time, and provides an explicit construction of the continuous time risk-neutral drift for both (discrete local) volatilities and spot. It also has a natural estimation strategy for its parameter space.

The core architecture follows the SANOS approach [6] which models the stock price as the product of a time-and-state discrete process X and a smoothing process Y, in our case Black & Scholes. The former is used to encode the option market using a discrete form of local volatilities as code book. That does not mean that the model is a local volatility model – we merely use the implied local volatilities as a convenient representation of the option surface. Compared to using implied volatilities the advantage is that mere positivity of the local volatilities is sufficient to ensure the the option market is free of static arbitrage at all times. We have shown in [6] that this representation can be fitted to real life market data very efficiently using linear programming.

The key contribution in this paper is showing how to construct a dynamic model of SANOS option call prices which are not only statically but also dynamically free of arbitrage. We show the theoretical construction of the corresponding risk-neutral measure $\mathbb{Q}$ and discuss existence of the respective martingale processes. We then discuss implementation and estimate our model on S&P Index Option data from January 2022 to August 2025. We fit our daily model parameters to observed bid/ask prices and then proceed to estimate both $\mathbb{Q}$ and its drift under $\mathbb{P}$. We show

[1] The model presented was trained for daily expiries for 25 business days and strikes between $[\exp(-2.5\sigma_{atm}\sqrt{t}), \exp(+1.5\sigma_{atm}\sqrt{t})]$.

that on this data set more than 90% of calibrated model prices fit within observed market bid/ask spreads even with the simplistic grid architecture we present here.

Our approach is closely related to the seminal but unpublished work [21]. In [10] the authors derive related conditions for continuous time local volatility models, but without a tangible implementation strategy. Our results can be seen as a joint extension of the two papers, plus the smoothing from our SANOS [6].

Related Work

Our work is further related to [11] where the authors solve the problem of finding a dynamically arbitrage-free evolution of the full option surface via products of indepdent Levy processes. While structurally very elegant option prices generated by products of independent increment processes do not typically fit well to equity index European option surfaces. Our approach also has the advantage of being numerically accessible. In particular, our model prices can be fit very efficiently within nearly all[2] of observed market bid/ask prices using the linear programming SANOS [6] calibration method.

The model presented here is distinct from our own DYSANOS [7]. There, we focused on learning realistic market dynamics for smooth statically arbitrage-free SANOS surfaces under $\mathbb{P}$. We did also construct an equivalent risk-neutral measure $\mathbb{Q}$ numerically using the *near-martingale measure* approach introduced in [8]. While this method is model-free and can in theory be applied to any option market model under $\mathbb{P}$, it is numerically heavy and in practise restricted to making only a fairly limited number of the overall tradable instruments dynamically arbitrage-free.[3] In the current work we cover all options with an explicit construction of the risk-neutral drift, which is far more efficient than the numerical near-martingale measure approach.

Hans comments

Maybe it's worth to state what we each think where we are and what the gaps are. Please comment–:

1. **DLV risk neutral diffusion and future densitioes** given a front density volatility: I think we are all clear in principle. I also noted you managed to prove existence which is really good. One quesition is what happens if $q \downarrow 0$... do we have an answer to that?

2. **DLV sampling spot** I suppose I was thinking we can sample from the front; that does not mean full step. See below. I think this way front density diffusion and spot are exactly in line. It does mean spot end ups at grid points; I suspect we can't get around that without a guanrateed arb-free call price interpolator.

3. **DLV sampling strategy** to preserve martingale property – testing that.

4. **DLV Increasing grid** I think works but I need to write it down properly ... **Magnus** ?

5. $\mathbb{P}$ **estimation** I have some ChatGPT code but nothing uber-smart. The main thing is that I think we can actually estime at the DLV vol-of-vols indepdenently of spot. Since we know its volatility ξ we know that the ¶ drift needs to be in $(\xi'\xi)^{-1}\xi$ times some vector of appropriate dimension.

[2] In below experiments, above 90%.
[3] In [7] we used 49 instruments per simulated date.

6. **DLV Smoothing** I am in the cheap camp: if X (the DLV process with all its option prices) is a margingale, then XY for Y an independent martingale is also martingale. If Y is (extended) Black & Scholes then I think it needs to be the same under ¶ and $\mathbb{Q}$. In general we can simply use a $\mathbb{Q}$ martingale and estimate the ¶ drift through X alone. Note that products with independent martingales create more of a "U" shape. I think **Blanka** you said you found a way to correlate X with Y?

Assume fixed expiry dates are $0 = T_0 < T_1 < \cdots < T_M$. Assume K are the strikes and that q_t^j is the density for T_j at time $t \leq T_j$. I think the core arguments are as follows:

2 Stochastic Smooth Discrete Local volatility

In this section we introduce the core model architecture for our *Stochastic Smooth Discrete Local Volatility (DLV) model*, SDLV. We follow the core SANOS [6] idea of writing our equity process S as product

$$S_t = X_t Y_t \tag{1}$$

of a stochastic discrete local volatility (SDLV) process X which is used to fit to current market condition and whose state parameters will have consistent $\mathbb{P}$ and $\mathbb{Q}$ dynamics, and an independent[4] and rather trivial smoothing processs Y whose dynamics under $\mathbb{P}$ and $\mathbb{Q}$ are the same.

The process Y will just be a Black & Scholes process with fixed volatility 5%. Given observed bid/ask market data it was shown in [6] how to fit a discrete local volatility surface in the presence of such process.

In the remainder of this section we focus on the SLDV process itself.

2.1 SDLV

The SLDV process will be defined on a *fixed* expiry and cash strike grid. It will not use a floating grid in its option market dynamics.[5] We will use "t" to refer to model time. We assume initial spot $X_0 = 1$.

For now let $0 = T_0 < T_1 < \cdots < T_M$ be the fixed model expiry dates, and assume that $K_0 < K_1 < \cdots < 1 < \cdots < K_{N+1}$ are the fixed model cash strikes, where K^0 and K^{N+1} are boundary strike which are far away from the inner strikes, c.f. [6].[6]

Assume that t is the current model time and let T_k be the front expiry such that $T_{k-1} \leq t < T_k$. Our model is given at any time t by the current spot X_t and future densities q_t^j, where $q_t^{j,i}$ represents the risk-neutral density of the i-th cash strike at future expiry T_j.

Such discrete densities have been shown in [9] to admit an equivalent representation via (squared) *discrete local volatilities* $\Sigma_t^{j,1}, \ldots, \Sigma_t^{j,N}$ in the following form: there is a tridiagonal

[4] Under both $\mathbb{P}$ and $\mathbb{Q}$.

[5] This is not to be confused to how the model parameters are estimated using historic market data. We explain estimation section **??**.

[6] In our experiments the inner strikes are given as range of strikes in from $K^1 = \exp(-2.5\sigma_{atm}(T_M)\sqrt{T_M})$ to $K^N = \exp(+1.5\sigma_{atm}(T_M)\sqrt{T_M})$ with boundaries set to $K^0 = K^1/2$ and $K^{N+1} = 3K^N/2$; see section **??** for specifics.

matrix Ω_j[7] for each $j > k$ such that

$$q_t^{j-1} = M_t^j q_t^j \ , \qquad M_t^j := I - \Omega_j S_t^j \ , \qquad S_t^j := \mathrm{diag}(0, \Sigma_t^{j,1}, \ldots, \Sigma_t^{j,N}, 0) \tag{2}$$

where M_t^j is an invertible tridiagonal M-matrix for every non-negative S_t^j. The key point is that q_t^j for $j > k$ are uniquely determined martingale densities for any given initial density q_t^k and the surface of non-negative (squared) discrete local vols.

2.2 Risk-Neutral Dynamics

Under a risk-neutral measure $\mathbb{Q}$ densities are martingales. We will now turn this property into conditions on dynamics of the (squared) discete local volatilities, and, jointly the front density and spot. Our approach follows in spirit that of [21] in the sense that we rely on the martingale property of a discrete set of tradable derivative assets. However, we reframe it in a much more accessible coordinate system.

Assume that W is a d-dimensional Brownian motion. We designate W^1 as the spot factor: spot is always driven by dW_t^1.

2.2.1 Future Densities

Each density coordinate is a martingale. We therefore specify the front-density diffusion as

$$dq_t^k = E_t^{k;1}\, dW_t^1 \ , \qquad E_t^{k;1} := \mathrm{diag}(q_t^k)\eta_t^{k;1} \ , \qquad E_t^{k;f} := 0 \quad (f = 2, \ldots, d),$$

where $E_t^{k;1} \in \mathbb{R}^{N+2}$ must satisfy $\mathbf{1}^\top E_t^{k;1} = 0$, so that q_t^k retains unit mass. The remaining Brownian factors do not directly move the front density. They drive the stochastic DLV surface, while $\xi_t^{j,\cdot,1}$ determines its leverage with spot. We do not specify η_t^j for $j > k$; the future density volatilities will be obtained recursively.

For future expiries T_j with $j > k$, assume the discrete local volatity squared is given as

$$\frac{d\Sigma_t^j}{\Sigma_t^j} = \mu_t^j\, dt + \xi_t^j\, dW_t,$$

where $\xi_t^j \in \mathbb{R}^{N\times d}$ is given and $\mu_t^j \in \mathbb{R}^N$ is the unknown risk-neutral drift. We write this as

$$dS_t^j = A_t^j\, dt + \sum_{f=1}^{d} B_t^{j;f}\, dW_t^f,$$

with $A_t^j := \mathrm{diag}(0, \Sigma_t^{j,1}\mu_t^{j,1}, \ldots, \Sigma_t^{j,N}\mu_t^{j,N}, 0)$ and $B_t^{j;f} := \mathrm{diag}(0, \Sigma_t^{j,1}\xi_t^{j,1;f}, \ldots, \Sigma_t^{j,N}\xi_t^{j,N;f}, 0)$. With the above

$$dM_t^j = -\Omega_j\, dS_t^j = -\Omega_j A_t^j\, dt - \sum_{f=1}^{d} \Omega_j B_t^{j;f}\, dW_t^f.$$

[7]The matrix $\Omega_j \in \mathbb{R}^{(N+2)\times(N+2)}$ is given as follows: let $\gamma_t^{\pm i} := \pm(K^i)^2(T_j - T_{j-1})/(K^{i+1} - K^{i-1})/(K^{i\pm1} - K^i)$ for $i = 1, \ldots, N$. Then Ω_j is the tridiagonal matrix with entries

$$(\Omega_j)_{i,i-1} = \gamma_t^{-i}, \quad (\Omega_j)_{i,i} = -(\gamma_t^{-i} + \gamma_t^{+i}), \quad (\Omega_j)_{i,i+1} = \gamma_t^{+i}, \quad i = 1, \ldots, N.$$

c.f. [6].

Setting $dq_t^j = \sum_f E_t^{j;f}\, dW_t^f$ for an unknown $E_t^{j;f}$ yields

$$dq_t^{j-1} = M_t^j\, dq_t^j + (dM_t^j) q_t^j + d[M^j, q^j]_t \tag{3}$$

$$= \sum_{f=1}^{d} \left(M_t^j E_t^{j;f} - \Omega_j B_t^{j;f} q_t^j \right) dW_t^f - \Omega_j \left(A_t^j q_t^j + \sum_{f=1}^{d} B_t^{j;f} E_t^{j;f} \right) dt\ . \tag{4}$$

Matching the diffusion coefficients gives

$$E_t^{j;f} = (M_t^j)^{-1} \left(E_t^{j-1;f} + \Omega_j B_t^{j;f} q_t^j \right) \in \mathbb{R}^{N+2} \quad \text{and} \quad \eta_t^{j,i;f} = \frac{E_t^{j,i;f}}{q_t^{j,i}}\ . \tag{5}$$

The drift must vanish, hence

$$\Omega_j \left(A_t^j q_t^j + \sum_{f=1}^{d} B_t^{j;f} E_t^{j;f} \right) = 0.$$

The restriction of Ω_j to vectors supported on the interior nodes is injective. Therefore the bracket vanishes node by node, giving

$$\mu_t^{j,i} = -\sum_{f=1}^{d} \xi_t^{j,i;f} \frac{E_t^{j,i;f}}{q_t^{j,i}} = -\sum_{f=1}^{d} \xi_t^{j,i;f} \eta_t^{j,i;f}, \qquad i = 1, \ldots, N. \tag{6}$$

Thus the prescribed front-density volatility and the prescribed DLV volatilities determine all future density volatilities and the risk-neutral DLV drifts by forward induction over the expiries.

2.2.2 Front Density and Spot

The remaining task is to specify both spot dynamics consistent with the front density and the respective front density volatilities.

Case $t = T_k - dt$

Assume we are at $t = T_k - dt$ and observe the front density q_t^k. Current spot is given as $S_t = K^\top q_t^k$. In order to sample a spot consistent with the discrete set of strikes, we can either draw a discrete sample or fit some other distribution exactly to the observed option prices (fitting to densities or probabilities will not work in general).

The simplest method is to draw spot from this discrete distribution using the spot factor W^1. Let F_t^k denote the conditional CDF which assigns probability $q_t^{k,i}$ to strike K_i, and let Φ denote the standard normal CDF. Then

$$S_{T_k} = (F_t^k)^{-1}(\Phi(Z))\,, \qquad Z := \frac{W_{T_k}^1 - W_t^1}{\sqrt{dt}} \sim N(0,1).$$

Correlation between spot and a future DLV coordinate is specified by that coordinate's loading $\xi_t^{j,i;1}$ on W^1.

Conditional on this definition of terminal spot, the front-density diffusion is exact if the Brownian increment is revealed progressively: define

$$Q_i := \sum_{\ell=0}^{i} q_t^{k,\ell}, \qquad a_i := \Phi^{-1}(Q_i), \qquad a_{-1} := -\infty, \quad a_{N+1} := +\infty,$$

and let ϕ be the standard normal density. For $t \le s < T_k$, set

$$z_i(s) := \frac{\sqrt{dt}\, a_i - (W_s^1 - W_t^1)}{\sqrt{T_k - s}}.$$

The time-s front density is the conditional probability

$$q_s^{k,i} := \mathbb{P}(S_{T_k} = K_i \mid \mathcal{F}_s) = \Phi(z_i(s)) - \Phi(z_{i-1}(s)). \tag{7}$$

It is therefore a martingale. Applying Itô's formula gives the exact diffusion

$$dq_s^{k,i} = E_s^{k,i;1}\, dW_s^1, \qquad E_s^{k,i;1} = \frac{\phi(z_{i-1}(s)) - \phi(z_i(s))}{\sqrt{T_k - s}}, \qquad t \le s < T_k, \tag{8}$$

with zero drift. Since $z_i(t) = a_i$, its initial value is

$$E_t^{k,i;1} = \frac{\phi(a_{i-1}) - \phi(a_i)}{\sqrt{dt}} \quad \text{hence} \quad \eta_t^{k,i;1} = \frac{\phi(a_{i-1}) - \phi(a_i)}{q_t^{k,i}\sqrt{dt}} \quad \text{for } i = 0, \dots, N+1. \tag{9}$$

while $E_t^{k,i;f} = \eta_t^{k,i;f} = 0$ for $f = 2, \dots, d$. More generally, $\eta_s^{k,i;1} = E_s^{k,i;1}/q_s^{k,i}$ and the other factor loadings remain zero. Consequently,

$$S_s = K^\top q_s^k = \mathbb{E}[S_{T_k} \mid \mathcal{F}_s], \qquad dS_s = K^\top E_s^{k;1}\, dW_s^1,$$

so spot and the front density are driven by the same Brownian factor. At expiry, q_s^k converges to the point mass associated with the sampled strike. The displayed coefficient $E_t^{k,i;1}$ is only the initial value of the exact, state- and time-dependent coefficient $E_s^{k,i;1}$; holding it constant over the whole interval would instead be an Euler approximation. Likewise, sampling S_{T_k} all at once without progressively revealing W^1 produces a direct jump to the terminal point mass rather than a continuous density diffusion.

Case $t \ll T_k$

The same construction gives a natural exact transition over any step which does not cross the front expiry. Let

$$h := T_k - t, \qquad 0 < \delta \le h, \qquad \Delta W_t^1 := W_{t+\delta}^1 - W_t^1.$$

Starting from q_t^k, define Q_i and a_i as above. After drawing $\Delta W_t^1 \sim N(0, \delta)$, set

$$q_{t+\delta}^{k,i} = \Phi\left(\frac{\sqrt{h}\, a_i - \Delta W_t^1}{\sqrt{h - \delta}}\right) - \Phi\left(\frac{\sqrt{h}\, a_{i-1} - \Delta W_t^1}{\sqrt{h - \delta}}\right), \qquad \delta < h. \tag{10}$$

The corresponding spot is not sampled independently but is defined by

$$S_{t+\delta} := K^\top q_{t+\delta}^k = \mathbb{E}[S_{T_k} \mid \mathcal{F}_{t+\delta}]. \tag{11}$$

Consequently

$$\mathbb{E}_t[q_{t+\delta}^k] = q_t^k, \qquad \mathbb{E}_t[S_{t+\delta}] = S_t$$

exactly. In general $S_{t+\delta}$ is not a grid strike; only at T_k does the density collapse to a unit vector and spot become the corresponding strike.

This update is recursively consistent. Indeed, the Gaussian boundaries calculated from the cumulative probabilities of $q_{t+\delta}^k$ are precisely the residual boundaries appearing in (10). The construction may therefore be restarted from $q_{t+\delta}^k$ without changing the joint law. It is a simplex-valued information bridge: conditional on the exact terminal Gaussian variable, W^1 is a Brownian bridge; conditional only on the terminal strike, it is Brownian motion conditioned to terminate in an interval, or equivalently a mixture of Brownian bridges.

2.2.3 Existence [TODO]

We have just described a set of risk-neutral model dynamics for a discrete process X_t whose future densities themselves are martingales driven by stochastic discrete local volatilities. Our construction is concise and lends itself to efficient numerical implementation and model parameter estimation.

[TODO] Existience

2.3 Sampling

In this section we discuss the sampling procedure for the discrete local volatility model. The final model is a product of this process and a simple Black & Scholes process with, say, volatility of 5%. It serves as a SANOS [6] smoother and ensures that the overall option price surface is smooth in strike.

On a time grid $t_n < t_{n+1}$ with $\delta_n = t_{n+1} - t_n$, one may proceed as follows.

First draw one Brownian vector $\Delta W_n \in \mathbb{R}^d$, using its first coordinate both in (10) and in every DLV coordinate with a non-zero loading $\xi^{j,i;1}$. This preserves the specified leverage. Update the front density by (10), and calculate spot from (11). This part of the update has no time-step bias, preserves positivity and unit mass, and is exactly martingale preserving.

For the DLV squared it is useful to work with $X_t^{j,i} := \log \Sigma_t^{j,i}$. Its Itô dynamics are

$$dX_t^{j,i} = \left(\mu_t^{j,i} - \frac{1}{2} \sum_{f=1}^{d} (\xi_t^{j,i;f})^2 \right) dt + \sum_{f=1}^{d} \xi_t^{j,i;f} \, dW_t^f .$$

Even the frozen-coefficient exponential update

$$\Sigma_{n+1}^{j,i} = \Sigma_n^{j,i} \exp\left\{ \left(\mu_n^{j,i} - \frac{1}{2} \sum_{f} (\xi_n^{j,i;f})^2 \right) \delta_n + \sum_{f} \xi_n^{j,i;f} \Delta W_n^f \right\} \tag{12}$$

is preferable to Euler integration of Σ: it cannot create a negative local variance. For state-dependent coefficients, a genuine Itô weak-order-two stochastic Runge–Kutta method in the log variables reduces the remaining time-step bias.

After updating Σ^j reconstruct the densities expiry by expiry by solving the tridiagonal systems

$$q_{n+1}^j = \left(I - \Omega_j S_{n+1}^j \right)^{-1} q_{n+1}^{j-1}, \qquad j = k+1, \ldots, M. \tag{13}$$

This enforces the DLV relation at every simulated state. Because the matrix is an M-matrix for non-negative Σ, the solve preserves positivity.

The drift condition (6) removes the infinitesimal drift of every density coordinate. It does not, by itself, make a generic finite-step discretisation exactly unbiased, since the map $\Sigma^j \mapsto (M^j)^{-1} q^{j-1}$ is nonlinear. Thus (12), or any ordinary higher-order SDE scheme, retains a time-discretisation bias which should vanish under step refinement.

Our experiments in **??** use this sampling scheme with antithetically paired Brownian increments.

Remark 2.1 (Alternative discrete-branch construction). If exact martingality under a chosen finite-step simulation law is required, it can instead be imposed directly. Choose symmetric

discrete shock branches $\Delta W_{n,b}$ with probabilities w_b and construct the front-density branches so that

$$\sum_b w_b q^k_{n+1,b} = q^k_n \ .$$

Parameterise each positive DLV update with an unknown drift correction $c^j_n \in \mathbb{R}^N$. For example, replace the log drift in (12) by $c^{j,i}_n$. For each expiry in turn, solve the nonlinear system

$$\sum_b w_b q^j_{n+1,b}(c^j_n) = q^j_n, \qquad q^j_{n+1,b}(c^j_n) = \left(I - \Omega_j S^j_{n+1,b}(c^j_n)\right)^{-1} q^{j-1}_{n+1,b}. \tag{14}$$

Only N coordinates are independent because mass and first moment are already preserved, matching the N drift corrections. Initialising the nonlinear solver with the continuous-time drift (6) gives a stable finite-step correction. Equation (14) makes the densities exactly martingales under the discrete branch law; symmetric or antithetic branches reduce variance but do not replace this condition.

An option market model reverses the usual direction of construction: instead of specifying a process for the underlying and deriving option prices from it, one models the code-book of prices directly, so that a calibration is a change of coordinates rather than an inverse problem. Two conditions must then be maintained at once. The state must remain free of *static* arbitrage, and its dynamics must admit one equivalent local martingale measure common to the underlying and to every traded option. The second is the option analogue of an HJM drift restriction: the volatility structure may be specified freely, and the drift is then determined. The lattice here is *finite*, finitely many strikes and finitely many expiry dates, which is what is traded.

What is already known. Schweizer and Wissel treat the problem in the continuum in two companion papers, one in each direction of the surface: [18] varies maturity at a single strike, and [17] varies strike at a single maturity. In each the code-book is chosen so that its positivity encodes absence of arbitrage, the drift restriction is derived, and existence is proved. The multi-strike multi-maturity case, which is ours, is stated as open in [17], with the expectation that a new parametrisation will be needed. Carmona and Nadtochiy [10] give a continuum local-volatility code-book with a drift restriction, but no existence theorem. On a finite lattice, and answering the multi-maturity question, stands Wissel [21]; his construction is the one this paper is closest to, and it is described separately below.

Modern approaches. The question was reopened from the data side, by the scarcity of option-market histories for training and evaluating hedging strategies [5]. Cohen, Reisinger and Wang [12] build neural stochastic differential equations on a floating lattice and penalise a least-squares residual in place of the drift restriction. [19] simulate an equity option book with a generative adversarial network whose state is the discrete local variance grid compressed by principal components, so that static arbitrage is absent by positivity of that grid; that is the code-book choice this paper inherits, and [20] extend it to a multi-asset universe. [7] decode statically arbitrage-free surfaces from a low-dimensional code, fit a linear state process to it, and test for dynamic arbitrage after the fact, reporting it in 12.5% of states at zero transaction cost.

None of the four enforces the drift restriction exactly, and that is where this paper starts: the cross-sectional constraints govern each date's surface and its terminal payoff cone, and leave the transition of a fixed contract between two dates unconstrained. The choice of grid is part of the same point. All four index their nodes relative to the spot rather than in fixed cash terms, and on such a grid a node is not one contract through time, so there is no single instrument whose price is required to be a martingale and nothing for an HJM-type restriction to constrain. Fixing the strikes in cash is what buys the restriction here.

The closest prior work. Wissel [21] works on a finite lattice of fixed cash strikes and fixed expiry dates, parametrises it by a price level together with positive local implied volatilities, derives the no-arbitrage drift restrictions, and proves constructive strong existence and uniqueness for a class of coefficients. Behind the front expiry his local implied variance is *exactly* the backward discrete local variance used here, an algebraic identity on the grid, and his drift restriction is, after a change of measure, exactly the one we state in Theorem 4.2. We claim no priority for it. That working paper appears never to have been published in archival form.

What this paper does

We keep the same lattice and change the chart: the state is the vector of gridded *densities* together with the discrete local variances, rather than prices together with quantiles of digitals. Three consequences follow, and they are the contributions.

1. **The spot is a read-off, and it settles exactly.** The spot is $S_t = K^\top q_t^j$ at every expiry j (Proposition 4.3), an identity of the chart holding at all t. At the front expiry Proposition 5.2 exhibits a family of front volatilities under which q^1 collapses onto a single node, so the spot lands on a strike and every gridded contract settles at its exact payoff (Section 5.2). That pinning is a property of the volatility chosen and not of the chart: by Remark 5.3 the terminal states the chart admits carry every spot in the grid range, whereas in [21] the pinning follows from the chart whatever the coefficients. We do not construct dynamics reaching an off-lattice terminal spot; that is left as future work.

2. **Existence, with no condition on the wings.** For every bounded and locally Lipschitz volatility structure b and any number of driving factors, the model has a unique strong solution on $[0, T]$ for every $T < T_1$, positive by construction; it extends to T_1 (Theorem 6.6(ii)) and, restarting at each expiry with the settling front, to the whole ladder $[0, T_M]$, unique given that front (Corollary 6.8). The quantity that must be controlled is $\eta = \psi/q$, the density volatility relative to the density, and its growth across expiries is bounded by the grid alone: Theorem 6.4 gives $\Theta_j \le N + 1$ uniformly in the state and in the volatility structure. *Wissel [21] must assume a bound of this kind; here it is a theorem.* The front expiry costs one hypothesis, and it is a mild one: η^1 must be integrable in t, not bounded, because η^j is affine in η^1 and an Itô equation needs an integrable drift rather than a bounded one. The front volatility used here diverges at rate $(T_1 - t)^{-p}$, and the two demands made of p leave a window: $p < 1$ is what Assumption 6.5 asks, while $p \ge \frac{1}{2}$ is what makes the front density collapse, so that every gridded contract settles at its exact payoff (Proposition 5.2).

3. **The drift is pointwise, and the count works out.** In density coordinates the drift equation is $N + 2$ conditions on N unknowns, and is consistent because the left null space of $\widehat{G}_j$ is exactly $\operatorname{span}\{\mathbf{1}, K\}$, conservation of mass and of the spot (Lemma 3.4). The state-independent factor then cancels from both sides and the drift is available nodewise, $\alpha^{j,i} = -\langle \eta^{j,i}, \beta^{j,i} \rangle$, with no matrix inversion and no rank condition.

Scope, and what is not claimed

The over-determination of the third contribution is not intrinsic: it is created by the choice of density coordinates and removed by the same choice, and in price coordinates the system is square and the question does not arise.

The existence statement is for a fixed finite grid. The constant is $(M-1)(N+1)$ and depends on the grid. The one hypothesis, Assumption 6.5, is a condition on the front volatility and not on the state, and it is sufficient rather than necessary.

3 The code-book

We assume deterministic forwards and discount factors, which reduces to a driftless spot with unit discount factors, cf. [4]. The spot at $t = 0$ is S_0.

The approach we are presenting relies on a combination of a time-and-state backbone process X and an additional smoothing process Y, following the [6] approach to smoothing. However, contrary to that model here we present a full stochastic local volatility model. The weight of the model is carried by the former and echos ideas of [21].

We will define the backbone process on a grid of fixed expiries $0 = T_0 < T_1 < \cdots < T_M$ and strikes $0 < K^0 < K^1 < \cdots < S_0 < \cdots < K^N < K^{N+1}$. We will use two time indices: t will denote model time and j will index fixed expires. We will use i to index strikes Contrary to [6] and similar to [21] expiries and strikes are not floating; they are fixed. The model will evolve along the grid. Specifically, at time t the future expiries in the model are those T_j such that $T_j > t$. We assume that the largest strike spacing is $h := \max_i(K^{i+1} - K^i)$.

3.1 Discrete Local Volatility Backbone

At time t we denote by $\Sigma_t^j = (\Sigma_t^{j,i})_{i=1,\dots,N}$ the squared discrete local volatility at the interior nodes $i = 1, \dots, N$ for expiry $T_j > t$. We let $\lambda = \log \Sigma$ throughout. Define

$$\omega^k := \frac{(K^k)^2}{K^{k+1} - K^{k-1}}, \qquad \iota^{k,-} := \frac{\omega^k}{K^k - K^{k-1}}, \qquad \iota^{k,+} := \frac{\omega^k}{K^{k+1} - K^k}, \qquad k = 1, \dots, N, \tag{15}$$

and then the tridiagonal matrix $A_j(\Sigma)$ with rows

$$\big(A_j(\Sigma)q\big)_i = \begin{cases} q^0 - \Delta_j \Sigma^1 \iota^{1,-} q^1, & i = 0, \\ -\Delta_j \Sigma^{i-1} \iota^{i-1,+} q^{i-1} + \big(1 + \Delta_j \Sigma^i (\iota^{i,-} + \iota^{i,+})\big) q^i - \Delta_j \Sigma^{i+1} \iota^{i+1,-} q^{i+1}, & 1 \le i \le N, \\ -\Delta_j \Sigma^N \iota^{N,+} q^N + q^{N+1}, & i = N+1. \end{cases} \tag{16}$$

(here q^i denotes the i-th component of a generic $q \in \mathbb{R}^{N+2}$). Up to notation this is the implicit forward operator of [9].

Let now X_t be the current spot price at time t for the backbone process X and let $q_t^0 \in \mathbb{R}^{N+2}$ the density

$$q_t^{0,i} = \frac{(X_t - K^{i-1})^+ - (X_t - K^i)^+}{K^i - K^{i-1}} - \frac{(X_t - K^i)^+ - (X_t - K^{i+1})^+}{K^{i+1} - K^i}$$

Then, the family $q_t = (q_t^j)_{j=0,\dots,M}$ starting in q_t^0 is then given iteratively as

$$A_j(\Sigma_t^j)\, q_t^j = q_t^{j-1} \tag{17}$$

Note that in (16) the entry multiplying $q_t^{i\pm1}$ carries $\Sigma_t^{i\pm1}$ and not Σ_t^i. The following was shown in [9]:

Theorem 3.1. *The family is then $q = (q_t^{j,i})$ with $q_t^j \in \mathbb{R}^{N+2}$ us a discrete risk-neutral density on the grid (T, K) in the sense that*

1. $q_t^{j,i} \ge 0$
2. $\sum_{i=0}^{N+1} q_t^{j,i} = 1$
3. $\sum_{i=0}^{N+1} K^i q_t^{j,i} = X_0$.

This means there is a discrete time-and-space process $X_t = (X_t^j)_{j=0,\dots,M}$ which takes values only in (T, K). Note that the model-time index here is j. Call prices for X_t for arbitrary $(\tau, k) \in \mathbb{R}^2_{\geq 0}$ are given by the linear functional

$$C_t(k,\tau) = \mathbb{E}\left[\left(X_t^j - k\right)^+\right] = \sum_{i=0}^{N+1} (K^i - k)^+ q_t^{j,i} \ . \tag{18}$$

with $j = \max\{j : T_j \leq \tau\}$.

Because (16) is affine in Σ we may write

$$A_j(\Sigma)\, q \ = \ q + G_j(q)\, \Sigma, \qquad G_j(q) \in \mathbb{R}^{(N+2)\times N}, \tag{19}$$

so that $G_j(q)$ collects the terms carrying Σ. The map G_j is linear in q and $G_j(q)\Sigma$ is linear in each argument separately; we use both readings. The following elementary observation is used constantly.

Lemma 3.2 (Factorisation)**.** *Column k of $G_j(q)$ involves q only through q^k. Consequently*

$$G_j(q) \ = \ \widehat{G}_j \,\mathrm{diag}(q^1, \dots, q^N), \tag{20}$$

where $\widehat{G}_j \in \mathbb{R}^{(N+2)\times N}$ has k-th column $\Delta_j\big(-\iota^{k,-}e_{k-1} + (\iota^{k,-} + \iota^{k,+})e_k - \iota^{k,+}e_{k+1}\big)$. Moreover $\widehat{G}_j$ has rank N, hence is injective.

Proof. Immediate from (16), whose terms carrying Σ^k are all proportional to q^k. For injectivity, suppose $\widehat{G}_j c = 0$ and let k be the smallest index with $c_k \neq 0$. Row $k-1$ receives a contribution from column k' only for $k' \in \{k-2, k-1, k\}$, and the first two carry $c_{k'} = 0$. So row $k-1$ reads $-\Delta_j \iota^{k,-} c_k = 0$, forcing $c_k = 0$, a contradiction. □

Note that $\widehat{G}_j$ depends only on the grid and on Δ_j, not on the state; the state enters (20) only through the diagonal factor.

Proposition 3.3 (The code-book)**.** *Fix the front density q^1, a probability vector on the grid with mean S_0, and let $j = 2, \dots, M$.*

(i) Given densities q^{j-1} and q^j with $q^j > 0$, the discrete local variance is uniquely determined by the linear system

$$G_j(q^j)\, \Sigma^j \ = \ q^{j-1} - q^j. \tag{21}$$

(ii) Conversely, given $\Sigma^j > 0$, the vector $q^j := A_j(\Sigma^j)^{-1} q^{j-1}$ is again a probability vector with mean S_0, and the gridded prices $(\pi^i)^\top q^j$ admit no static arbitrage.

(iii) A gridded surface is free of static arbitrage, butterflies and calendar spreads together, if and only if it arises as in (ii) *from some $\Sigma \geq 0$, and it lies in the* interior *of the arbitrage-free set if and only if $\Sigma > 0$.*

Proof. (i) is (19), with uniqueness from Lemma 3.2. For (ii), $A_j(\Sigma^j)^{-1}$ is non-negative with unit column sums and inherits the martingale property [9], attributed there to Andreasen and Huge [1]; non-negativity excludes butterfly arbitrage, and q^j is then a mean-preserving spread of q^{j-1}, so Jensen's inequality excludes calendar arbitrage. (iii) is [9], after [3]: absence of arbitrage is equivalent to non-negative Gamma and Backward-Theta at every node, and Σ is their ratio. Both conditions are nodewise, and $\Sigma^{j,i} = 0$ exactly when Backward-Theta vanishes. □

What everything else rests on is that the admissible set is a product of half-lines, one per node, so it has no faces in the sense that a polytope does, and $\lambda = \log \Sigma$ maps its interior diffeomorphically onto $\mathbb{R}^{N(M-1)}$.

3.2 Two conservation laws

The whole structure of the drift restriction follows from one observation about the left null space of G_j. Both identities below are used by [9], where they hold by construction for the backward operator. What is not there, because nothing in that paper needs it, is that $\operatorname{span}\{\mathbf{1}, K\}$ is the *entire* left null space. That is a rank statement, and it is what makes the drift system of Section 4 square.

Lemma 3.4 (Left null space)**.** *For every q and every j,*

$$\mathbf{1}^\top G_j(q) = 0 \qquad \text{and} \qquad K^\top G_j(q) = 0,$$

where $\mathbf{1} = (1, \dots, 1)$ and $K = (K^0, \dots, K^{N+1})$. Consequently $A_j(\Sigma)$ conserves both total mass and the spot, for any Σ. Combined with Lemma 3.2, the left null space of $\widehat{G}_j$ is exactly $\operatorname{span}\{\mathbf{1}, K\}$.

Proof. Fix an interior index k and collect the three entries of $G_j(q)$ in column k: from (16) they are $\Delta_j(\iota^{k,-} + \iota^{k,+})q^k$ in row k, $-\Delta_j \iota^{k,+} q^k$ in row $k+1$, and $-\Delta_j \iota^{k,-} q^k$ in row $k-1$. Their sum is 0, giving the first identity. Weighting by the node values,

$$\begin{aligned}\Delta_j q^k \Big[K^k\big(\iota^{k,-} + \iota^{k,+}\big) - K^{k+1}\iota^{k,+} - K^{k-1}\iota^{k,-}\Big] &= \Delta_j q^k \Big[\iota^{k,-}(K^k - K^{k-1}) - \iota^{k,+}(K^{k+1} - K^k)\Big] \\ &= \Delta_j q^k \big[\omega^k - \omega^k\big] = 0\end{aligned}$$

by (15), giving the second. The conservation statements follow from (19): $\mathbf{1}^\top A_j(\Sigma) q = \mathbf{1}^\top q$ and $K^\top A_j(\Sigma) q = K^\top q$. Since $\widehat{G}_j$ has rank N and lives in $\mathbb{R}^{(N+2)\times N}$, its left null space is two-dimensional, hence exactly $\operatorname{span}\{\mathbf{1}, K\}$. □

Remark 3.5 (Counting). For $q > 0$, $G_j(q)$ has $N+2$ rows and N columns and rank N. Hence the $N+2$ martingale conditions per expiry contain exactly two redundancies, conservation of mass and of the spot, and the remaining N conditions match the N unknowns of Σ^j. The system is square on the quotient.

4 The drift restriction

Theorem 4.1 (Martingale property)**.** *Assume that there is a measure $\mathbb{Q}$ such that each $q_t^{j,i}$ is a martingale. Then X_t and all options written on it are martingales.*

Under $\mathbb{Q}$ every gridded instrument price is a martingale in t. Since the gridded payoffs π^i together with the unit payoff span $\mathbb{R}^{N+2}$, and $\mathbf{1}^\top q^j = 1$, this is equivalent to each density vector being a martingale,

$$\mathrm{d}q_t^j = \psi_t^j \,\mathrm{d}W_t, \qquad \psi_t^j \in \mathbb{R}^{(N+2)\times d}. \tag{22}$$

This is the only assumption behind the drift restriction, and it carries no regularity. Each q^j is a fixed linear combination of traded butterflies, so (22) is the martingale representation theorem applied to traded prices, and nothing has to commute with a stochastic integral. This is the substantive advantage of a discrete code-book over its continuum analogue, where the corresponding step requires differentiating a stochastic field in strike and maturity and must be licensed by standing regularity conditions. Theorem 4.2 below needs only $q^j > 0$.

Theorem 4.2 (Drift restriction)**.** *Let* $\mathrm{d}\Sigma_t^j = \alpha_t^j \mathrm{d}t + \beta_t^j \mathrm{d}W_t$ *and* $q^j > 0$*. Under Assumption 4.1, for each* $j = 2, \dots, M$*,*

$$A_j(\Sigma^j)\,\psi^j \;=\; \psi^{j-1} - G_j(q^j)\,\beta^j, \tag{23}$$

marched forward in j *from* ψ^1*, the volatility of the front density, which is given by its chart (Section 5). With the relative density volatility*

$$\eta^{j,i} \;:=\; \frac{\psi^{j,i}}{q^{j,i}}, \qquad i = 0, \dots, N+1, \tag{24}$$

the drift is determined node by node,

$$\alpha^{j,i} \;=\; -\langle \eta^{j,i}, \beta^{j,i} \rangle \qquad \Longleftrightarrow \qquad \mathrm{d}\lambda^{j,i} = \Big[-\langle \eta^{j,i}, b^{j,i} \rangle - \tfrac{1}{2} |b^{j,i}|^2 \Big] \mathrm{d}t + b^{j,i}\,\mathrm{d}W_t, \tag{25}$$

where $\lambda = \log \Sigma$ *and* $b = \beta / \Sigma$*.*

Proof. By Lemma 3.2, $G_j(q)\Sigma = \widehat{G}_j(\Sigma \circ q)$, where $\Sigma \circ q := (\Sigma^i q^i)_{i=1}^N$ is the entrywise product on the interior nodes. The code-book therefore reads

$$q_t^j + \widehat{G}_j\, p_t^j = q_t^{j-1}, \qquad p^{j,i} := \Sigma^{j,i} q^{j,i}, \quad i = 1, \dots, N,$$

with $\widehat{G}_j$ constant. Itô's product rule for the scalar processes $\Sigma^{j,i}$ and $q^{j,i}$ gives

$$\mathrm{d}p^{j,i} = \Big[q^{j,i} \alpha^{j,i} + \langle \psi^{j,i}, \beta^{j,i} \rangle \Big] \mathrm{d}t + \big(\Sigma^{j,i} \psi^{j,i} + q^{j,i} \beta^{j,i} \big) \mathrm{d}W_t, \tag{26}$$

with $\psi^{j,i}, \beta^{j,i} \in \mathbb{R}^{1 \times d}$ the rows at node i. Hence $\mathrm{d}q^j + \widehat{G}_j\,\mathrm{d}p^j = \mathrm{d}q^{j-1}$, and by uniqueness of the decomposition of a continuous semimartingale the $\mathrm{d}W$ and $\mathrm{d}t$ terms match separately. The $\mathrm{d}W$ terms give $\psi^j + \widehat{G}_j \big(\Sigma^j \circ \psi^j + q^j \circ \beta^j \big) = \psi^{j-1}$, factor by factor, which is (23) by (19). Since q^j and q^{j-1} are driftless by Assumption 4.1, the $\mathrm{d}t$ terms give

$$\widehat{G}_j \Big(q^j \circ \alpha^j + \textstyle\sum_f \psi^{j,\cdot,f} \circ \beta^{j,\cdot,f} \Big) = 0.$$

Since $\widehat{G}_j$ is injective (Lemma 3.2) the bracket vanishes, $q^{j,i} \alpha^{j,i} + \langle \psi^{j,i}, \beta^{j,i} \rangle = 0$ node by node, which is (25) after dividing by $q^{j,i} > 0$. The logarithmic form is Itô applied to $\lambda = \log \Sigma$, using $\alpha / \Sigma = -\langle \eta, b \rangle$. □

Only the interior components of η enter (25), since Σ lives on the interior nodes; the boundary components are needed in Section 6.

Equation (23) is not ours. Compared with the pricing recursion $A_j(\Sigma^j) q^j = q^{j-1}$ it is *the same operator with one added source*, the structure of Carmona and Nadtochiy [10]. The drift restriction above is theirs in the continuum, and Wissel's [21] on this lattice; we claim no priority for either.

What the fixed lattice changes is threefold. Their restriction carries a term in the derivative of the code-book along time to maturity; a lattice of fixed expiry dates has no such derivative, and here that term is absent. The relation is algebraic rather than differential, so neither the high-order regularity their statement requires nor the singular behaviour as time to maturity vanishes that it is built to accommodate can arise here, the front slab having positive width. And the discrete system is over-determined, $N + 2$ conditions on N unknowns, which is what the counting of Remark 3.5 squares.

Two features of (25) are worth drawing out. The first is its shape. It is the exact analogue of the HJM restriction $\alpha = \sigma \int \sigma$: the drift of the log discrete local variance is minus the covariance of its own volatility with the volatility of the log density, plus an Itô convexity term. Where HJM's derived object is an integral of the volatility across maturities, ours is η, obtained from b by the single forward recursion (23). The second is the cost. Once η is known no matrix is inverted: no Jacobian of the pricing map is ever formed and no rank condition on it appears. And since ψ is affine in β and α is bilinear in (β, ψ), the drift is a quadratic polynomial in β, as in HJM; Theorem 6.4 is what rules out the corresponding finite-time explosion.

The driftless point is the classical model: if $\beta \equiv 0$ and $\psi^1 = 0$ then $\psi \equiv 0$ and $\alpha \equiv 0$, so the discrete local variance surface is frozen, and with $\psi^1 \neq 0$ the densities still move, driven by the front alone, with $\alpha \equiv 0$ persisting.

Proposition 4.3 (Spot consistency)**.** *The spot is $S_t = K^\top q_t^j$, the same for every j by Lemma 3.4. Since $\mathbf{1}^\top A_j = \mathbf{1}^\top$, $K^\top A_j = K^\top$ and $\mathbf{1}^\top \widehat{G}_j = K^\top \widehat{G}_j = 0$, equation (23) gives*

$$\mathbf{1}^\top \psi^j = \mathbf{1}^\top \psi^1 = 0 \qquad \text{and} \qquad K^\top \psi^j = K^\top \psi^1 = \sigma_t^S S_t \quad \text{for every } j.$$

So the spot's volatility is the single row $K^\top \psi^1 \in \mathbb{R}^{1\times d}$, it is automatically the same at every expiry, and no further condition links the spot to the surface. Taking $\psi^1 = 0$ freezes the spot and recovers $K^\top \psi^j = 0$.

Proposition 4.3 is the discrete form of Carmona and Nadtochiy's spot consistency condition: the spot and its volatility are both *read off* the surface rather than modelled, $S = K^\top q^j$ and $\sigma^S S = K^\top \psi^j$ for any j. If $K^0 = 0$, the read-off is the zero-strike call, $S_t = C_t(0, T_j)$ for every j, as in [21] and in the multi-strike model of [17].

The drift restriction is used below through the following chain, which is what a model actually evaluates:

$$(\psi^1, b) \xrightarrow{(23)} \psi^2, \ldots, \psi^M \xrightarrow{\eta = \psi/q} \alpha, \tag{27}$$

a forward march of d tridiagonal solves per expiry, $O(NMd)$ in total. The volatility structure is the only input; the drift is computed, never fitted.

5 The spot and the front expiry

The spot is not a state variable. By Proposition 4.3 it is the read-off $S_t = K^\top q_t^j$, the same scalar at every expiry. What has to be specified is the front density q^1, the initial condition of the recursion of Proposition 3.3; everything at $j \geq 2$ follows from it and from the volatility structure.

5.1 The front chart

The front density must be strictly positive and sum to one. Both are automatic in the chart

$$q^{1,i} = \frac{e^{z^i}}{\sum_k e^{z^k}}, \qquad \mathrm{d}z = m\,\mathrm{d}t + \sum_{f=1}^{d} \nu^f\,\mathrm{d}W^f, \tag{28}$$

with $z \in \mathbb{R}^{N+2}$ unconstrained. The chart is invariant under $z \mapsto z + c\mathbf{1}$, so a shift of ν along $\mathbf{1}$ leaves q^1 unchanged and only the q^1-centred part of ν is identified; write $\mathrm{dev}^f := \nu^f - (q^1)^\top \nu^f\,\mathbf{1}$

for it. The input ν is free, and plays the role b plays at $j \geq 2$. Itô's formula for a function of a vector process shows that each $q^{1,i}$ is a martingale exactly when

$$m^i = -\tfrac{1}{2} \sum_f \big(\mathrm{dev}^{i,f}\big)^2 \tag{29}$$

up to a multiple of $\mathbf{1}$. Furthermore we obtain

$$\psi^{1,\cdot,f} = q^1 \circ \mathrm{dev}^f, \qquad \eta^{1,\cdot,f} = \mathrm{dev}^f, \qquad \mathbf{1}^\top \psi^{1,\cdot,f} = 0, \tag{30}$$

with $\circ$ the entrywise product. The same computation appears in [13] and as the stochastic replicator equation [15].

Remark 5.1 (Resolution at the front expiry). Among grid measures with mean S the least dispersed is the pair of nodes straddling it, so

$$\mathrm{Var}_{q^1}(K) \;\geq\; (S - K^i)(K^{i+1} - S), \qquad K^i \leq S < K^{i+1}, \tag{31}$$

which is at most $\frac{1}{4}h^2$. The front implied volatility therefore has the floor

$$\frac{\sqrt{(S - K^i)(K^{i+1} - S)}}{S\sqrt{T_1 - t}} \;\leq\; \frac{h}{2S\sqrt{T_1 - t}},$$

largest at the midpoint of a cell and zero when the spot sits on a node, which is exactly what Proposition 5.2 forces at T_1.

5.2 Terminal settlement: pinning the spot to a strike

The chart constrains q^1 at no time $t < T_1$ beyond positivity and mass, so it imposes no terminal condition. Wissel's front parametrisation fixes the terminal state whatever the coefficients [21]; here the terminal state is a modelling choice. We take the front volatility proportional to the distance from the spot, in one factor,

$$\eta_t^{1,i} = \frac{c_t\,(K^i - S_t)}{h}, \qquad c_t = a_1 \tau^{-p} \text{ near } T_1, \quad \tau = T_1 - t,\ a_1 > 0. \tag{32}$$

Only the behaviour near T_1 matters: on $[0, T_1 - \varepsilon]$ the front volatility may be any bounded, locally Lipschitz choice, and (32) need only hold on $[T_1 - \varepsilon, T_1)$. It is already centred, since $\sum_i q^{1,i}(K^i - S_t) = 0$. Its coefficients are polynomial in q^1 and bounded on $[0, T]$ for every $T < T_1$, so there is a unique strong solution on $[0, T_1)$,

$$q_t^{1,i} = q_0^{1,i} \exp\Big(\int_0^t \eta^{1,i}\,\mathrm{d}W - \tfrac{1}{2}\int_0^t (\eta^{1,i})^2 \mathrm{d}s \Big), \qquad \|\eta_t^1\|_\infty \leq c_t\, w/h, \quad w := K^{N+1} - K^0. \tag{33}$$

Proposition 5.2 (The front collapses onto a node, and settles exactly)**.** *If $p \geq \frac{1}{2}$ then $q_{T_1}^1 := \lim_{t \uparrow T_1} q_t^1 = e_k$ for a random index k, almost surely, so $S_{T_1} = K^k$ and*

$$\sum_i q_{T_1}^{1,i} (K^i - K^n)^+ = (S_{T_1} - K^n)^+ \qquad \textit{for every } n. \tag{34}$$

If $p < 1$ then Assumption 6.5 holds on $[0, T_1]$. The window $p \in [\frac{1}{2}, 1)$ gives both.

Proof. Each $q^{1,i}$ is a bounded martingale, so $q^1_t \to q^1_{T_1}$ almost surely, and the variance $V_t := \sum_i q^{1,i}_t (K^i - S_t)^2$ converges to that of $q^1_{T_1}$. By (30) and (32) the volatility of $S = K^\top q^1$ is $\sum_i K^i q^{1,i} \eta^{1,i} = c_t V_t / h$ on $[T_1 - \varepsilon, T_1)$. Since S is a martingale with values in $[K^0, K^{N+1}]$,

$$\mathbb{E} \int_{T_1-\varepsilon}^{T_1} \frac{c_t^2 V_t^2}{h^2} \, \mathrm{d}t = \mathbb{E}\big[(S_{T_1} - S_{T_1-\varepsilon})^2\big] \le w^2,$$

so the integral is finite almost surely. On the event $\{V_{T_1} > 0\}$ we would have $V_t \ge V_{T_1}/2$ near T_1, and the integral would diverge, because $\int^{T_1} c_t^2 \, \mathrm{d}t = a_1^2 \int^{T_1} \tau^{-2p} \, \mathrm{d}t = \infty$ for $2p \ge 1$. Hence $V_{T_1} = 0$ almost surely: $q^1_{T_1}$ is a probability vector on distinct points with zero variance, that is e_k with $K^k = S_{T_1}$, and (34) reads $(K^k - K^n)^+ = (S_{T_1} - K^n)^+$. Finally η^1 is bounded on $[0, T_1 - \varepsilon]$ and $\int_{T_1-\varepsilon}^{T_1} \|\eta^1_t\|_\infty \mathrm{d}t \le (w/h) \int_{T_1-\varepsilon}^{T_1} c_t \, \mathrm{d}t < \infty$ when $p < 1$, by (33). □

The terminal spot is therefore a grid point, as in [21]; the difference is that here it is chosen, by (32), and there it is forced by the parametrisation.

Remark 5.3 (What the lattice could carry, and does not). At expiry each butterfly price $q^{j,i}$ is worth its payoff: for $S_{T_j} \in [K^k, K^{k+1}]$ the second difference defining $q^{j,i}$ telescopes to $\varpi(S_{T_j})$, the barycentric weights of the spot on the two nodes straddling it. Since $u \mapsto (u - K^n)^+$ is affine between nodes,

$$\sum_i \varpi^i(s) \, (K^i - K^n)^+ \;=\; (s - K^n)^+ \qquad \text{for every } s \in [K^0, K^{N+1}] \text{ and every model strike } K^n, \tag{35}$$

so $\varpi(s)$ is an admissible terminal state that settles every gridded contract exactly, also for s strictly inside a cell. The chart does not force the pinning; the choice (32) does, because its terminal variance must vanish. A front volatility that drives q^1 to $\varpi(s)$ with s inside a cell is left as future work (Section 7).

6 The wing constant, and existence

The log discrete local variance $\lambda = (\lambda^2, \dots, \lambda^M)$, $\lambda^j = \log \Sigma^j$, ranges over all of $\mathbb{R}^{N(M-1)}$, so positivity is automatic; the front density has its own chart (28). By Theorem 4.2 the model is the stochastic differential equation

$$\mathrm{d}\lambda_t = \Big[-\langle \eta_t, b_t \rangle - \tfrac{1}{2} |b_t|^2 \Big] \mathrm{d}t + b_t \, \mathrm{d}W_t, \tag{36}$$

whose only non-elementary coefficient is $\eta = \psi / q$ from (23) and (24). Its denominator is small in the wings, and the question is whether η can blow up there.

The argument has the structure of Wissel's proof of his Theorem 3.5 [21]: log coordinates, induction over expiries, and local existence followed by a bound that rules out explosion. Its one non-standard ingredient is an a priori bound, independent of the state, on the relative volatility of the calendar spread at the second expiry. Wissel could not obtain this bound in general and imposes his condition (3.24) instead (his Remark 1 following the proof, p. 18); in our coordinates the quantity is $\eta + b$. We obtain it from an exact identity for the volatility recursion and a known inequality for inverses of M-matrices.

6.1 The volatility recursion

Write $P_j := A_j(\Sigma^j)^{-1}$, so that $q^j = P_j q^{j-1}$, and P when j is fixed.

Corollary 6.1 (Positivity, unit column sums, absorbing ends)**.** *$P \geq 0$ entrywise, $\mathbf{1}^\top P = \mathbf{1}^\top$, so $P_{im} \in [0,1]$, and the outer columns of P are e_0 and e_{N+1}.*

Proof. By (16), A_j has non-positive off-diagonal entries, and in each column the diagonal entry exceeds the sum of the moduli of the off-diagonal ones by exactly one; so A_j is a non-singular M-matrix and $P \geq 0$ [2]. The column sums follow from $\mathbf{1}^\top A_j = \mathbf{1}^\top$ (Lemma 3.4), and the outer columns of A_j are e_0 and e_{N+1}. The first two statements are also Theorem 3.3 of [9], attributed there to [1]. □

That is all that is needed about P itself. What we want next is a recursion for η alone: one step per expiry, each adding a controlled amount, so that the bound telescopes along the ladder. Dividing (23) by q^j gives

$$\eta^{j,i} \;=\; \frac{(P\psi^{j-1})_i}{q^{j,i}} \;-\; \frac{\big(P\,G_j(q^j)\beta^j\big)_i}{q^{j,i}}, \tag{37}$$

the first term carrying η^{j-1} and the second carrying b^j.

The first is an average. Writing $\psi^{j-1} = q^{j-1} \circ \eta^{j-1}$, its weights $P_{ik}q^{j-1,k}/q^{j,i}$ are non-negative because $P \geq 0$ and sum to one because $q^j = Pq^{j-1}$, so it contributes at most $\|\eta^{j-1}\|_\infty$; they are the matrix $\mathcal{A}$ of (42).

The source is the difficulty. Since G_j is a second difference, row i carries the neighbouring densities $q^{j,i\pm1}$ as well as $q^{j,i}$, so dividing by $q^{j,i}$ leaves the ratios

$$\frac{q^{j,i-1}}{q^{j,i}} \qquad \text{and} \qquad \frac{q^{j,i+1}}{q^{j,i}}, \tag{38}$$

which are unbounded in the wings, where the density decays fastest; nothing in (23) bounds them.

They are, however, an artefact of writing the row out. Since $\mathbf{1}^\top G_j(q) = 0$ by Lemma 3.4, row i of $G_j(q)\Sigma^j$ is the net outflow of node i, and by (19) that net is

$$G_j(q^j)\,\Sigma^j \;=\; A_j(\Sigma^j)q^j - q^j \;=\; q^{j-1} - q^j, \tag{39}$$

which carries no ratios at all: the individual fluxes are wing quantities, their net is not. For $b^j \equiv c$ this settles the matter, since $\beta^j = c\,\Sigma^j$ and $Pq^{j-1} = q^j$ give

$$\frac{\big(P\,G_j(q^j)\beta^j\big)_i}{q^{j,i}} = c\,\frac{\big(q^j - Pq^j\big)_i}{q^{j,i}} = c\Big(1 - \frac{(Pq^j)_i}{q^{j,i}}\Big),$$

the two ratios of (38) having collapsed into the single propagation ratio $(Pq^j)_i/q^{j,i}$. A general b only modulates the density. Let $E \in \mathbb{R}^{N\times(N+2)}$ restrict to the interior nodes, so $EE^\top = I_N$, and note that $A_j = I + \widehat{G}_j\,\mathrm{diag}(\Sigma^j)E$. Writing $E^\top(q \circ b)$ for the vector equal to $q^i b^i$ at the interior nodes and zero at the outer ones, (20) and the commuting of diagonal matrices give

$$G_j(q)\beta^j = \widehat{G}_j\,\mathrm{diag}(\Sigma^j)E\,E^\top(q \circ b^j) = \big(A_j(\Sigma^j) - I\big)E^\top(q \circ b^j), \tag{40}$$

again the operator's own increment, applied to $E^\top(q \circ b^j)$ in place of q. The proposition is what this amounts to.

Proposition 6.2. *With $b^j = \beta^j/\Sigma^j$ and $\circ$ the entrywise product,*

$$\psi^j \;=\; P\,\psi^{j-1} \;-\; (I-P)\,E^\top\big(q^j \circ b^j\big), \tag{41}$$

equivalently, with the entrywise non-negative

$$\mathcal{A} := \operatorname{diag}(q^j)^{-1} P \operatorname{diag}(q^{j-1}), \qquad \mathcal{B} := \operatorname{diag}(q^j)^{-1} P\,E^\top \operatorname{diag}(Eq^j),$$

$$\eta^j \;=\; \mathcal{A}\,\eta^{j-1} \;+\; \big(\mathcal{B} - E^\top\big) b^j, \tag{42}$$

where $\mathcal{A}\mathbf{1} = \mathbf{1}$, so the transport is an averaging operator, and the i-th row of $\mathcal{B}$ sums to $(Pq^j)_i/q^{j,i} - \mathbf{1}_{\{i\in\{0,N+1\}\}}$.

Proof. By (23), $\psi^j = P\psi^{j-1} - P\,G_j(q^j)\beta^j$. Applying P to (40) at $q = q^j$ and using $PA_j = I$ gives $P\,G_j(q^j)\beta^j = (I-P)E^\top(q^j \circ b^j)$, which is (41). Dividing it by $q^{j,i}$ gives (42): the rows of $\mathcal{A}$ sum to $(Pq^{j-1})_i/q^{j,i} = 1$, and for $\mathcal{B}$ the terms $k \in \{0, N+1\}$ missing from $(Pq^j)_i$ vanish for interior i and equal $q^{j,i}$ at $k = i$ for outer i, by Corollary 6.1. □

So η^j is an average of η^{j-1} plus a source in b^j, and the march across expiries is additive.

6.2 The wing constant

That single surviving ratio, $(Pq^j)_i/q^{j,i}$, is what has to be bounded. Since $q^j = Pq^{j-1}$ it is

$$\frac{(Pq^j)_i}{q^{j,i}} = \frac{(P^2q^{j-1})_i}{(Pq^{j-1})_i}, \tag{43}$$

so what is needed is how much larger than P the square P^2 can be entrywise; and since both sides are linear and non-negative in q^{j-1}, the case $q^{j-1} = e_m$ shows that an entrywise bound is not merely sufficient but necessary. Write

$$R^{j,i} := \mathbf{1}_{\{1\le i\le N\}} + \frac{(Pq^j)_i}{q^{j,i}} - \mathbf{1}_{\{i\in\{0,N+1\}\}}, \qquad \Theta_j := \max_i R^{j,i}, \qquad \Xi_j := \sum_{r=1}^{N} P_{rr} \le N, \tag{44}$$

and $H_j := \max_i|\eta^{j,i}|$. Here $R^{j,i}$ is the row bound of (42) at node i, and Θ_j, the *wing constant*, is its size across the slab. The theorem shows Θ_j is bounded by the grid alone, uniformly in the state; the bound rests on a single inequality for inverses of M-matrices.

Lemma 6.3 (Squared propagator)**.** *For all nodes i, r, m,*

$$P_{ir}\,P_{rm} \;\le\; P_{im}\,P_{rr}, \tag{45}$$

and consequently, for all i, m,

$$\big(P^2\big)_{im} \le \begin{cases} \Xi_j\,P_{im}, & 1 \le i \le N, \\ (1+\Xi_j)\,P_{im}, & i \in \{0, N+1\}. \end{cases} \tag{46}$$

Proof. P is the inverse of an M-matrix by Corollary 6.1, and every inverse M-matrix is a path product matrix, $a_{ij}a_{jk} \le a_{ik}a_{jj}$ for all i, j, k [14, Theorem 4.5]; (45) is the case $(i, j, k) = (i, r, m)$. Summing it over r gives (46): an outer r contributes only if $r = i$, since column r of P is e_r, and then contributes P_{im}. □

Theorem 6.4. *For every $\Sigma \in (0,\infty)^N$,*

$$H_j \;\le\; H_{j-1} + \Theta_j \|b^j\|_\infty, \tag{47}$$

and

$$\Theta_j \le 1 + \Xi_j \le N+1, \qquad \Theta^* := \sup_{\Sigma} \sum_{j=2}^{M} \Theta_j \le (M-1)(N+1). \tag{48}$$

Proof. Since $q^j = Pq^{j-1} \ge 0$, Lemma 6.3 gives $(Pq^j)_i = \sum_m (P^2)_{im} q^{j-1,m} \le \Xi_j\, q^{j,i}$ for interior i and $\le (1+\Xi_j) q^{j,i}$ for outer i, so $R^{j,i} \le 1+\Xi_j$ in both cases. Finally, $\mathcal{A} \ge 0$ with $\mathcal{A}\mathbf{1} = \mathbf{1}$ gives $\|\mathcal{A}\eta^{j-1}\|_\infty \le H_{j-1}$, while row i of $(\mathcal{B} - E^\top) b^j$ has modulus at most $\|b^j\|_\infty R^{j,i}$ by the row sums of Proposition 6.2. □

6.3 Existence

Iterating (47) from the front slice,

$$\sup_{j,i} |\eta_t^{j,i}| \;\le\; \|\eta_t^1\|_\infty + (M-1)(N+1)\, \|b\|_\infty, \tag{49}$$

and iterating (41) shows that η^j is affine in η^1, with coefficients depending on λ only through $P_2, \dots, P_j$ and $q^2, \dots, q^j$. The only t on the right of (49) is inside η_t^1, which enters linearly, so it has to be integrable in time rather than bounded.

Assumption 6.5 (Integrable front volatility)**.** For a horizon $T \le T_1$,

$$\int_0^T \|\eta_t^1\|_\infty \,\mathrm{d}t \;<\; \infty \qquad \text{almost surely.} \tag{50}$$

Theorem 6.6 (Existence)**.** *Let the front volatility $\nu = \nu(t,z)$ in* (28) *and the volatility $b = b(t,z,\lambda)$ be locally Lipschitz in (t,z,λ) on $[0,T_1)$, with $|b| \le B$, and ν bounded on $[0,T]$ for every $T < T_1$.*

(i) For every $T < T_1$ the front chart (28)–(29) *together with* (36) *has a unique strong solution (z,λ) on $[0,T]$, and*

$$0 \;<\; \inf_{t\le T} \min_{j,i} \Sigma_t^{j,i} \;\le\; \sup_{t\le T} \max_{j,i} \Sigma_t^{j,i} \;<\; \infty \qquad \textit{almost surely.} \tag{51}$$

(ii) If moreover Assumption 6.5 holds with $T = T_1$, then λ_t has a finite limit λ_{T_1} as $t \uparrow T_1$, and (51) *holds with $T = T_1$.*

No condition is imposed on Σ, on the wings, or on the number of factors.

Proof. (i) Fix $T < T_1$ and let B_ν bound $|\nu|$ on $[0,T]$. The coefficients are locally Lipschitz: $q^1 = \mathrm{softmax}(z)$ and $\Sigma = e^\lambda$ are positive, so each A_j is a non-singular M-matrix and everything except ν and b is smooth in (z,λ). Adding time as a state variable to cover the time dependence, [16, Theorem V.38, p. 303] gives a unique solution up to an explosion time ζ, with $\limsup_{t\to\zeta} \|(z_t,\lambda_t)\| = \infty$ on $\{\zeta < \infty\}$. The drift of λ is bounded by (49), since $\|\eta^1\|_\infty = \|\mathrm{dev}\|_\infty \le 2B_\nu$, and the drift m of z is bounded by $2B_\nu^2$; let D bound both. On $[0, \zeta\wedge T)$ the drift part of (z,λ) is therefore at most DT, and the noise part, having integrand bounded by $B_\nu + B$, is a martingale bounded in L^2, hence convergent as $t \to \zeta\wedge T$. So (z,λ) stays bounded

on $[0, \zeta \wedge T)$, which rules out $\zeta \le T$. The solution exists on $[0,T]$, λ is bounded there, and (51) follows from $\Sigma = e^{\lambda}$.

(ii) By (i) the solution exists on $[0,T_1)$. By (49) the drift of λ is bounded by $g(t) := \big(\|\eta_t^1\|_\infty + (M-1)(N+1)B\big)B + \frac{1}{2}B^2$, which is integrable on $[0,T_1)$ by Assumption 6.5, so the drift part of λ converges as $t \uparrow T_1$. The noise part $\int_0^t b\,\mathrm{d}W$ has integrand bounded by B, so it is a martingale bounded in L^2 on $[0,T_1)$ and converges as well. Hence λ_t has a finite limit, λ is bounded on $[0,T_1]$, and (51) holds with $T = T_1$. Nothing is claimed for z, which may run off at T_1 as the front collapses. □

The settling front (32) is bounded on every $[0,T]$ with $T < T_1$ by (33), and satisfies Assumption 6.5 with $T = T_1$ for $p < 1$ (Proposition 5.2). So for $p \in [\frac{1}{2}, 1)$ the code-book reaches its first expiry with Σ bounded away from 0 and ∞ and every gridded contract settling at its payoff.

6.4 Beyond the first expiry

At T_1 the front expires and the next expiry becomes the front. The settling of Proposition 5.2 makes this switch clean: the spot sits on a node, and the new front density is a column of the propagator. Indeed, if $q^l_{T_l} = e_k$ then the code-book $A_{l+1}(\Sigma^{l+1})\,q^{l+1} = q^l$ gives

$$q^{l+1}_{T_l} = P_{l+1}\,e_k, \qquad P_{l+1} := A_{l+1}\big(\Sigma^{l+1}_{T_l}\big)^{-1}, \tag{52}$$

the spread of a unit mass at the settled spot by one step of the code-book. To restart the front chart we need this density to be strictly positive.

Lemma 6.7 (The new front)**.** *Let $P = A_j(\Sigma^j)^{-1}$ with $\Sigma^j \in (0,\infty)^N$. If k is an interior node, then $(Pe_k)_i > 0$ for every node i and $K^\top P e_k = K^k$. If $k \in \{0, N+1\}$, then $Pe_k = e_k$.*

Proof. Order the nodes with the interior first. The outer columns of A_j are e_0 and e_{N+1} (Corollary 6.1), so

$$A_j = \begin{pmatrix} A_{II} & 0 \\ A_{BI} & I \end{pmatrix}, \qquad P = \begin{pmatrix} A_{II}^{-1} & 0 \\ -A_{BI}A_{II}^{-1} & I \end{pmatrix}.$$

By (16), A_{II} is tridiagonal with strictly negative off-diagonal entries, hence an irreducible non-singular M-matrix, so $A_{II}^{-1} > 0$ entrywise [2]. The block $-A_{BI} \ge 0$ has a positive entry in the row of K^0 (column K^1) and in the row of K^{N+1} (column K^N), so the outer entries of an interior column of P are positive as well. The identity $K^\top P = K^\top$ follows from $K^\top A_j = K^\top$ (Lemma 3.4), and the outer columns are Corollary 6.1. □

Corollary 6.8 (Existence on the whole ladder)**.** *Let b be bounded and locally Lipschitz. On each interval $[T_{l-1}, T_l]$, $l = 1, \dots, M$, let the current front density q^l follow the chart* (28) *with the settling volatility* (32)*, where $\tau = T_l - t$ and $p \in [\frac{1}{2}, 1)$. Then the model has a solution on $[0, T_M]$, unique given these choices, such that*

(i) every gridded price is a continuous martingale up to its expiry;

(ii) at every T_l the spot sits on a node K^{k_l} and every contract expiring at T_l settles at its payoff;

(iii) if k_l is interior, the model continues on $[T_l, T_{l+1}]$ from the finite state $z = \log q^{l+1}_{T_l}$ and $\lambda^{l+2}, \dots, \lambda^M$ at T_l; if k_l is an outer node, the spot is absorbed there: $q^j_t = e_{k_l}$ for all $j > l$ and $t \ge T_l$.

Proof. By induction on l. On $[0, T_1]$ this is Proposition 5.2 with Theorem 6.6, whose part (ii) also gives the limit λ_{T_1}, finite. Suppose the solution exists on $[0, T_l]$ with λ_{T_l} finite and $q^l_{T_l} = e_k$.

If k is interior, (52) and Lemma 6.7 give a strictly positive new front, so $z = \log q^{l+1}_{T_l}$ is finite, and $K^\top q^{l+1}_{T_l} = K^k = S_{T_l}$, so the spot is continuous. On $[T_l, T_{l+1}]$ the ladder is that of Sections 4–6 with front q^{l+1} and expiries $T_{l+1} < \cdots < T_M$, started from the $\mathcal{F}_{T_l}$-measurable state $(z, \lambda^{l+2}, \ldots, \lambda^M)$. Theorem 6.6 applies with this $\mathcal{F}_{T_l}$-measurable initial condition, the solution of [16, Theorem V.38] depending continuously on its starting point, and with Proposition 5.2 and part (ii) of that theorem gives a unique strong solution on $[T_l, T_{l+1}]$ that settles at T_{l+1} with $\lambda_{T_{l+1}}$ finite. The densities q^j, $j > l+1$, are given by the same recursion on both sides of T_l with continuous inputs, so they are continuous at T_l.

If k is an outer node, Lemma 6.7 gives $q^j_{T_l} = e_k$ for every $j > l$. The constant process $q^j_t \equiv e_k$ satisfies every recursion, since $A_j e_k = e_k$, and is a martingale, so all prices are constant and settle at their payoffs; we freeze λ at λ_{T_l}, on which no price then depends.

Uniqueness on each interval gives uniqueness on $[0, T_M]$. □

7 What is open

1. *An off-lattice terminal spot.* By Remark 5.3 the chart represents every $s \in [K^0, K^{N+1}]$ at expiry, and the state that does so is $\varpi(s)$, carried by the two nodes straddling s. Whether a front volatility exists driving q^1 to $\varpi(s)$ for s strictly inside a cell, rather than to a single node as (32) does, is open. The obstruction is that exact settlement forces $q^1_{T_1} = \varpi(S_{T_1})$ and hence $\int_0^{T_1} (\eta^{1,i})^2 \mathrm{d}t < \infty$ for both surviving nodes, which by the argument of Proposition 5.2 they cannot both satisfy when the volatility is proportional to distance from the spot. Until it is answered the model of Proposition 5.2 pins, as [21] does, and his question of a front that does not pin the spot remains open.

2. *The continuum limit.* The constant $(M-1)(N+1)$ grows with the grid. Whether $\sup_N \Theta_j < \infty$ on a class of realistic surfaces, which is what a continuum statement would need, is open.

8 Implementation Commets

Data and estimation. The implementation uses the daily 25-expiry by 200-strike data set. The historical DLV-squared innovations are aligned locally at fixed cash strike and fixed expiry over each one-day pair. Across different day pairs the surface and its support are allowed to float with the market. There are 710 training pairs through 31 December 2024. We use the full DLV state rather than the affine-200 state: the respective 2025 out-of-sample residual root mean squares are 0.256 and 0.423. Gaussian smoothing with widths one expiry and two strike nodes is applied before a rank-eight functional principal-component fit; the retained factors explain 72.88% of the smoothed innovation variance.

The historical spot shock is calculated exactly as proposed above,

$$Z_t = \Phi^{-1}\big(F^1_t(S_{t+1}/S_t)\big)\,,$$

where F^1_t is the fitted one-day front distribution including the independent 5% Black smoothing process. Its fitted mean and standard deviation are 0.0290 and 1.0026. The first DLV factor is

oriented to this score. Under $\mathbb{Q}$ the DLV drift is recomputed from (6) at every numerical substep. Under $\mathbb{P}$ we impose

$$\mu^{\mathbb{P}} = \mu^{\mathbb{Q}} + \xi\lambda$$

and project the empirical conditional DLV drift onto this attainable subspace. The statewise projection explains 62.20% of the weighted drift energy; a single constant λ used for simulation explains 42.15%. The independent 5% Black process is assigned zero risk premium and is therefore identical under $\mathbb{P}$ and $\mathbb{Q}$.

Numerical update. The front density is advanced with the exact Gaussian information bridge and the same first Brownian factor that drives spot leverage. Gaussian interval probabilities in the upper tail are evaluated as differences of survival probabilities, avoiding subtraction of two CDF values close to one. At the end of a day the expired front is sampled directly from the density at the start of that day using the aggregate first-factor increment. This is algebraically equivalent to recursively revealing the bridge, while avoiding accumulation of floating-point error in the settlement law.

DLV squared is evolved in log coordinates with eight uniform intraday substeps. The drift and front-density volatility are recomputed at each substep and the future density tower is reconstructed with (13). The ordinary frozen log-Euler drift increment $x = \mu\delta$ is used whenever $|x| \le 0.5$. One simulation path exposed a density of order 10^{-24} for which freezing the relative drift produced $x \simeq 2490$ and numerical overflow. Only for such exceptional increments we use the tail safeguard

$$x \longmapsto \operatorname{sgn}(x)\,(0.5 + \log(1 + |x| - 0.5))\,.$$

It leaves ordinary steps unchanged and has the same infinitesimal limit as the stated SDE. We do not clip DLV squared or alter the SANOS density recursion.

Several alternatives were rejected during implementation. An additive Euler update of the densities can leave the positive convex-order cone, and a simple antithetic recentering has the same problem. A balanced explicit–implicit DLV drift multiplier removed the overflow but introduced measurable bias on ordinary paths. A grid refined in squared remaining time did not improve a matched experiment. These failures are numerical rather than changes to the continuous-time drift identity.

CPU validation. A coupled 512-path, 25-day $\mathbb{P}/\mathbb{Q}$ pilot produced no non-finite values. Mass and forward errors were below 1.3×10^{-13}. The final risk-neutral audit used 100,000 paths, eight substeps and all 200 strikes. It tested 65,650 daily density returns, 65,000 daily latent-option returns, 5,050 density-to-expiry returns, 5,000 option-to-expiry returns including the Black smoothing process, and all 25 daily spot returns. The run had no non-finite values; its maximum mass and forward errors were 2.61×10^{-13} and 2.58×10^{-13}.

For rare claims, a Gaussian statistic based only on predictable variance can be misleading. In one reproduced example the exercise count was 96 against 96.44 expected, but one exceptionally large payoff gave a predictable- variance score of 9.18 while the realised studentised score was 0.98. Accordingly, the audit retains the analytically known conditional variance but uses the larger of predictable and empirical standard errors for inference. All expiry-density, expiry-option, daily-density and daily-spot statistics were below five standard errors, with maxima 3.53, 2.83, 3.83 and 2.27.

Two of the 63,810 statistically identifiable daily latent-option coordinates remained above five standard errors. Their worst statistic was 6.21, but the corresponding mean return was

only -5.27×10^{-8}. Thus the implementation is stable and the residual bias is economically tiny, but this experiment does not claim exact martingality for an arbitrary finite time step. This is consistent with the discussion following (13): the continuous-time drift condition removes the infinitesimal drift, not the nonlinear finite-step bias. Exact martingality under a specified discrete shock law would require solving the branchwise nonlinear correction (14).

A Model and DLV factor visualizations

A.1 Consistent Drift

Figure 1 illustrates the fitted measure change using four matched Model D paths. Solid and dashed paths of the same colour use identical Gaussian draws under $\mathbb{Q}$ and $\mathbb{P}$. Solid and dashed lines of the same colour use the same Gaussian draws under $\mathbb{Q}$ and $\mathbb{P}$. Besides spot, the illustration reports ATM one- and five-day implied volatility and the central log-strike skew

$$\text{Skew}_\tau = \frac{\sigma_{\text{imp}}(K = Z_t e^h, \tau) - \sigma_{\text{imp}}(K = Z_t e^{-h}, \tau)}{2h}, \qquad h = 0.01. \tag{53}$$

The paths illustrate the fitted measure change; four paths are not a physical model validation.

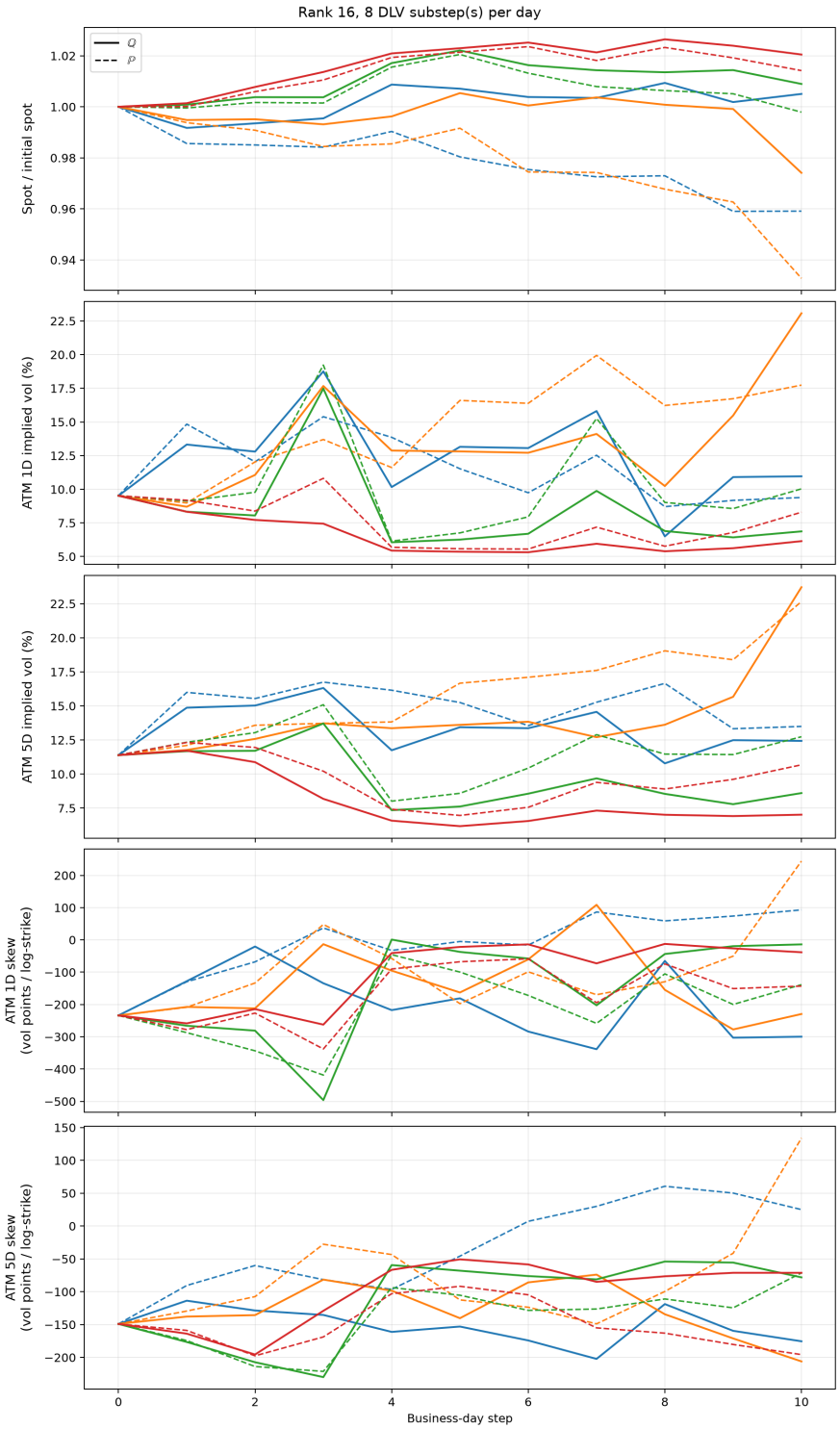


Figure 1: Matched $\mathbb{P}/\mathbb{Q}$ paths for Model D: rank 16 with eight Brownian substeps per day.

A.2 PCA Summary

The remaining figures visualize the factor systems estimated in Section **??**. Figure 2 is the baseline risk-neutral implied-volatility surface used for all factor illustrations.

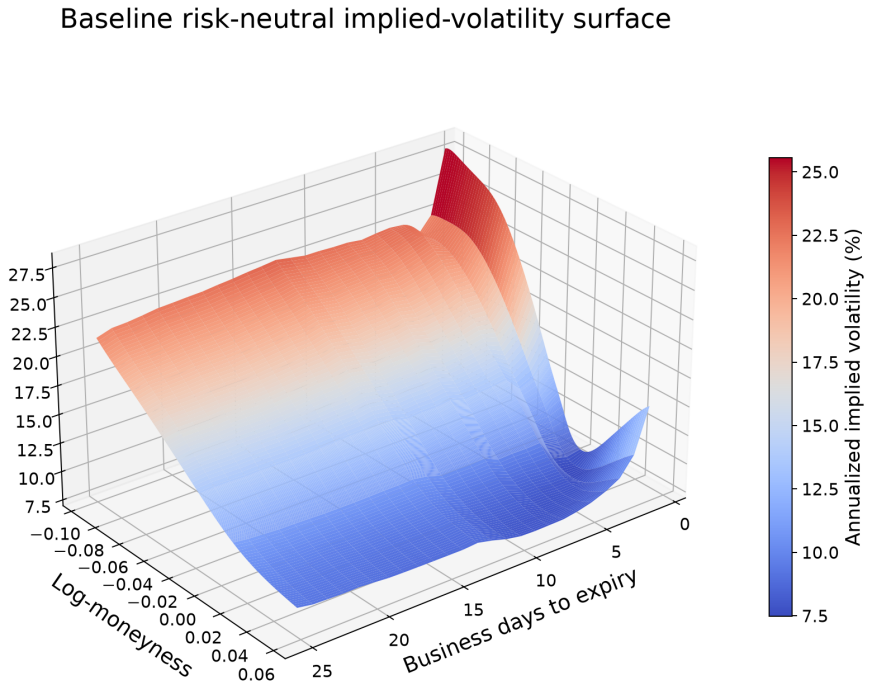


Figure 2: Baseline annualized implied volatility on the fitted log-moneyness and business-days-to-expiry grid.

Figure 3 reports individual and cumulative PCA power. The experiments retain either eight or 16 factors; the first rank above 90% is 26.

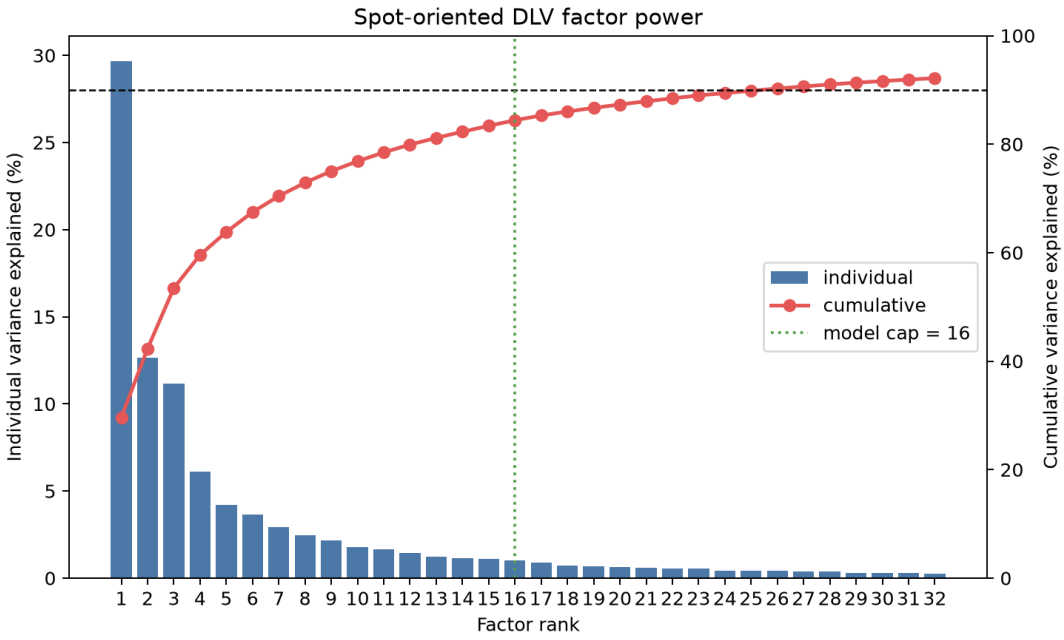


Figure 3: Individual and cumulative variance explained by the spot-oriented DLV factor system. The vertical line is the rank-16 model cap; 26 factors are required to exceed 90%.

The rank-16 fit appends PCs 9–16 to the rank-8 fit. A one-standard-deviation shock is the diffusion-only log-DLV displacement $\pm\xi_f\sqrt{1/252}$. The figures display its central implied-volatility response

$$\frac{1}{2}\left[\sigma_{\text{imp}}(\log\Sigma+\xi_f/\sqrt{252})-\sigma_{\text{imp}}(\log\Sigma-\xi_f/\sqrt{252})\right]. \tag{54}$$

These shock directions are the same under $\mathbb{P}$ and $\mathbb{Q}$: an equivalent measure change alters drift, not diffusion. Each panel deliberately retains its own vertical scale.

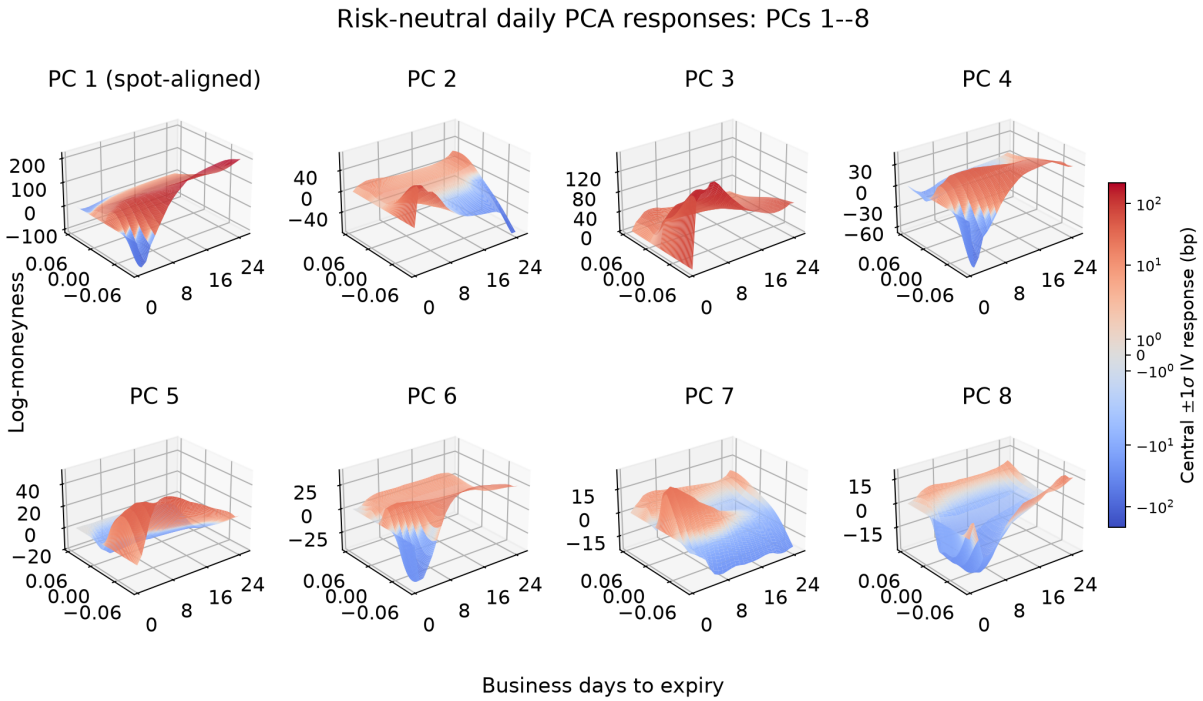


Figure 4: Risk-neutral one-standard-deviation implied-volatility responses for PCs 1–8. PC 1 is aligned to the front-spot score.

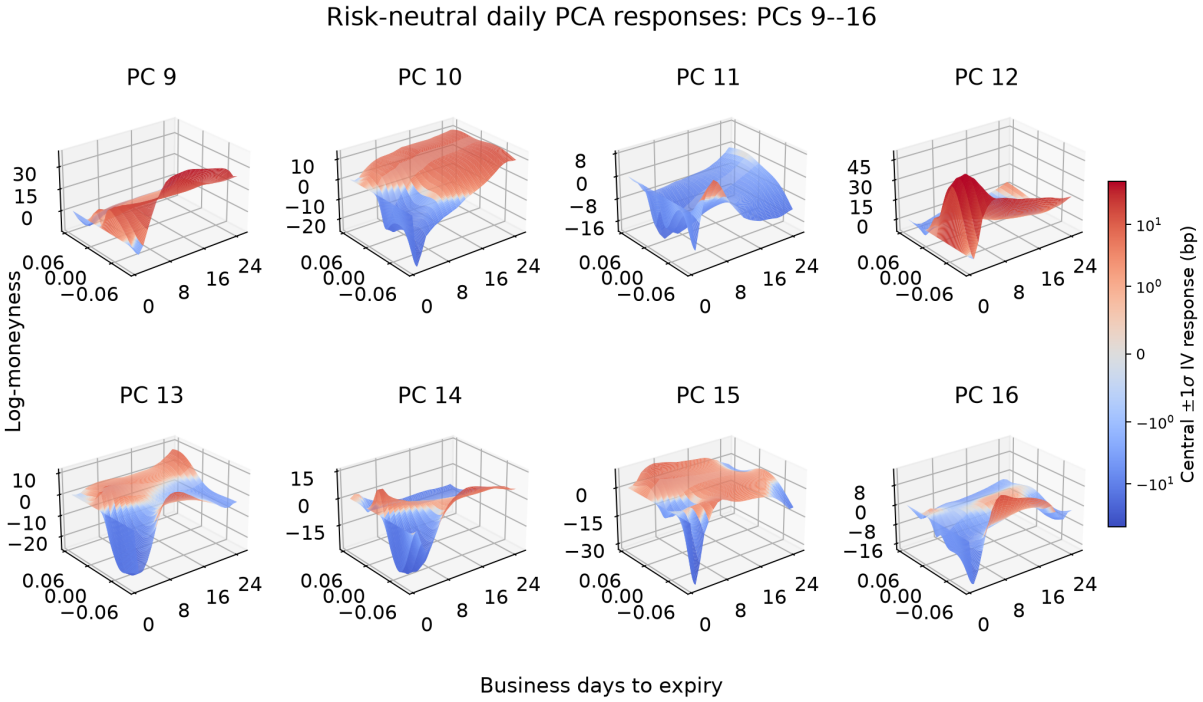


Figure 5: The additional one-standard-deviation implied-volatility responses, PCs 9–16, in the nested rank-16 model.

B Market-data and calibration pipeline

This appendix records the data and numerical pipeline behind Section **??**. It follows the broader SPX pipeline described in [7], restricted here to the short-dated fixed-expiry SANOS experiment. The complete transformation is

$$\begin{aligned}\text{raw IvyDB quotes} &\longrightarrow \text{daily cleaning and parity-forward enrichment}\\ &\longrightarrow \text{arbitrage-free SANOS eligibility surface}\\ &\longrightarrow \text{European short-horizon quote selection}\\ &\longrightarrow \text{daily 25-by-200 calibration tensor}\\ &\longrightarrow \text{direct or affine end-to-end fit.}\end{aligned}$$

B.1 Raw data and cleaning

The source is OptionMetrics IvyDB US data accessed through WRDS. For each eligible SPX market date we load the index level and previous index level, option strikes and calendar expiries, call and put bid and ask quotes, volume, open interest, AM/PM settlement flags, and the zero curve. The retained sample runs from 3 January 2022 through 29 August 2025 and contains 891 valid full-day NYSE sessions.

The raw cleaning step removes known bad ticker–date observations, missing or invalid quotes, invalid settlement flags, same-day expiries, irregular expiries, and quotes whose bid does not exceed 0.011. Business days and year fractions to expiry are then computed; an AM-settled expiry is treated as 0.2 of a business day earlier than its corresponding PM expiry.

For each expiry we recompute the forward from put–call parity rather than use the delayed IvyDB forward table. With discount factor D, define the discounted-forward bounds at a strike K by

$$G^{\mathrm{ask}}(K) = C^{\mathrm{ask}}(K) - P^{\mathrm{bid}}(K) + DK, \qquad G^{\mathrm{bid}}(K) = C^{\mathrm{bid}}(K) - P^{\mathrm{ask}}(K) + DK.$$

Their midpoint is averaged across strikes with inverse-spread weights,

$$\widehat{G} = \frac{\sum_K w_K \left[G^{\mathrm{ask}}(K) + G^{\mathrm{bid}}(K)\right]/2}{\sum_K w_K}, \qquad w_K = \frac{1}{\max\{G^{\mathrm{ask}}(K) - G^{\mathrm{bid}}(K), 10^{-3}\}}.$$

The resulting forward is used to construct the tightest call-equivalent quote, forward-normalized strikes and prices, implied volatilities, and Greeks. An expiry must contain at least ten strikes with a traded call or put and at least four strikes with both sides traded. The normalized spread must exceed 10^{-6} and retained implied volatilities must lie between 0.01 and 5.

B.2 Static preprocessing and quote selection

Before applying the model-specific horizon, the pipeline retains quotes with vega per unit total volatility of at least 10^{-3}, log volume and log open interest of at least one, and at least ten surviving options per expiry. The nominal per-day quote cap is 100,000 and is therefore non-binding here.

For data-quality control and to construct a stable ATM curve, every daily book is passed through the linear SANOS preprocessing of [6]. The strike grid is extended, each expiry is fitted inside the quoted bid–ask intervals, negligible-density redundant strikes are removed, and the joint surface fit enforces both butterfly and calendar-spread constraints. The smoothness parameter is

0.25. Expiries and strikes for which this arbitrage-free auxiliary fit is unavailable are not passed to the final calibration.

This auxiliary SANOS surface determines eligibility, the ATM term structure, and the floating model support. It is not the calibration target: the JAX fits in Section **??** use the original retained call and put bid–ask intervals. No linear-programming state or fitted midpoint is supplied as a target.

The final selection keeps European quotes no farther than 25 business days from the observation date, excludes the 0.2-business-day slice, and retains the standardized log-moneyness interval $[-2.5, 1.5]$. This leaves 2,192,458 quotes, between 1,590 and 5,096 per date.

B.3 Daily calibration tensor

The model expiry grid is

$$T_j = \frac{j}{252}, \qquad j = 1, \ldots, 25.$$

For each date, 100 interior-strike candidates are generated from the front ATM scale and 100 from the 25-day ATM scale. They are merged with a minimum log-spacing safeguard to form one 200-node support shared across all model expiries on that date. The support floats from date to date. Two boundary nodes, equal to one half of the lowest inner strike and one and a half times the highest inner strike, complete the 202-node density grid.

Each retained market quote stores its call/put flag, normalized strike, bid, ask, midpoint, business days to expiry, adjacent model-expiry indices, and linear interpolation weight. The contemporaneous SPX level is retained as a reference spot. The immutable calibration tensor pads each date to a multiple of 128 quotes and carries a Boolean mask, so padding never contributes to the objective.

At valuation time the DLV recursion produces the 25 discrete densities. The fixed 5% Black factor convolves each discrete density into a smooth call-price curve, and the two adjacent expiry curves are linearly interpolated to the market quote's exact expiry. Puts follow from put–call parity.

B.4 Optimization and retained artifacts

The direct and affine parameterizations are those in Section **??**. Direct states are initialized from the date-specific ATM term structure and are constant across strike at initialization. Affine means use the average of these initial states; codes and loadings receive small centered Gaussian initial values. DLV squared is bounded between 10^{-4} and 9, with a differentiable barrier near the upper bound.

The loss in (**??**) is differentiated over every retained quote. Quote blocks of size 128 are used only to limit accelerator memory: their gradients are summed before each Adam update, so this is an exact full-sample gradient rather than stochastic minibatching. Each fit uses 300 initial epochs with a cosine learning-rate schedule from 2×10^{-2} to 10^{-3}, followed by a 700-epoch continuation from 5×10^{-3} to 5×10^{-5}. Checkpoints store either all daily direct states or the affine mean, loadings, and daily codes. The retained reports recompute every quote from the final checkpoint and produce the statistics in Table **??**.

References

[1] Jesper Andreasen and Brian Norsk Huge. Volatility interpolation. Available at SSRN 1694972, 2010. Also published in Risk, March 2011.

[2] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*, volume 9 of *Classics in Applied Mathematics*. SIAM, 1994. Reprint of the 1979 Academic Press edition.

[3] Hans Buehler. Expensive martingales. *Quantitative Finance*, 6(3):207–218, 2006.

[4] Hans Buehler. Volatility and dividends - volatility modelling with cash dividends and simple credit risk. `https://papers.ssrn.com/abstract=1141877`, 2008.

[5] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19:1271–1291, 2019. SSRN working paper version: doi:10.2139/ssrn.3120710 (2018).

[6] Hans Buehler, Blanka Horvath, Anastais Kratsios, Yannick Limmer, and Raeid Saqur. Sanos - smooth arbitrage-free non-parametric option surfaces. `https://arxiv.org/abs/2601.11209`, Februar 2026.

[7] Hans Buehler, Blanka Horvath, and Anastasis Kratsios. Dysanos generative dynamic smooth arbitrage-free non-parametric option surfaces, 2026.

[8] Hans Buehler, Phillip Murray, Mikko S. Pakkanen, and Ben Wood. Deep hedging: learning to remove the drift. *Risk*, February 2022.

[9] Hans Buehler and Evgenii Ryskin. Discrete local volatility for large time steps (short version). Technical report, SSRN, 2016. Short version. An extended version also exists: SSRN 2642630 (2015), doi:10.2139/ssrn.2642630.

[10] René Carmona and Sergey Nadtochiy. Local volatility dynamic models. *Finance and Stochastics*, 13:1–48, 2009.

[11] René Carmona and Sergey Nadtochiy. Tangent lévy market models. *Finance and Stochastics*, 16(1):63–104, 2012.

[12] Samuel N. Cohen, Christoph Reisinger, and Sheng Wang. Arbitrage-free neural-sde market models, 2021. arXiv:2105.11053.

[13] Damir Filipović, Lane P. Hughston, and Andrea Macrina. Conditional density models for asset pricing. *International Journal of Theoretical and Applied Finance*, 15(1):1250002, 2012.

[14] Charles R. Johnson and Ronald L. Smith. Inverse M-matrices, II. *Linear Algebra and its Applications*, 435(5):953–983, 2011.

[15] Panayotis Mertikopoulos and Aris L. Moustakas. The emergence of rational behavior in the presence of stochastic perturbations. *The Annals of Applied Probability*, 20(4):1359–1388, 2010.

[16] P. E. Protter. *Stochastic Integration and Differential Equations*. Springer, 2 edition, 2005.

[17] Martin Schweizer and Johannes Wissel. Arbitrage-free market models for option prices: The multi-strike case. *Finance and Stochastics*, 12:469–505, 2008.

[18] Martin Schweizer and Johannes Wissel. Term structures of implied volatilities: Absence of arbitrage and existence results. *Mathematical Finance*, 18:77–114, 2008.

[19] Magnus Wiese, Lianjun Bai, Ben Wood, and Hans Buehler. Deep hedging: Learning to simulate equity option markets, 2019. NeurIPS 2019 Workshop on Robust AI in Financial Services.

[20] Magnus Wiese, Ben Wood, Alexandre Pachoud, Ralf Korn, Hans Buehler, Phillip Murray, and Lianjun Bai. Multi-asset spot and option market simulation, 2021. arXiv:2112.06823.

[21] Johannes Wissel. Arbitrage-free market models for option prices. NCCR FINRISK Working Paper 428, NCCR FINRISK, ETH Zürich, 2007.